Ep 8 · Fast & Cheap LLM Apps — Cheat Sheet

AI System Design · AI Engineering with Ajit · github.com/byruajit/ai-system-design

MEASURE FIRST

Budget every stage at p95: gateway · embed · search · rerank · **TTFT** · generation. Users judge by **time to first token**: stream.

COST LEVERS (BIGGEST FIRST)

1. **Route** easy → small model
2. **Fewer chunks** (rerank to 3)
3. **Prompt caching**: stable prefix first
4. **Exact + semantic cache** (safely)
5. Concise output, `max_tokens`
6. **Batch API** for offline jobs
7. **Self-host** past break-even

SAFE SEMANTIC CACHE

Risk	Guard
Personal data	Never cache "my/me" questions; scope keys
Near-miss	**Slot guard**: numbers & qualifiers match
Stale	Store doc versions; invalidate on change
Permissions	ACL scope in the cache key

Tune the threshold on labelled pairs: ship **zero wrong hits**.

SELF-HOSTING CHECKLIST

vLLM/TGI continuous batching · quantise only after evals · autoscale on queue depth · keep 1 warm replica · overflow to an API

METRICS

TTFT p95 · total p95 · tokens/request · cost/request · cache hit **and wrong-hit** rate · GPU utilisation

SAY THIS

"Streaming fixes perception; routing and chunk count fix the bill; the cache must be scoped, guarded and versioned."

Northwind Bank is fictional. Prices are illustrative tiers — check current provider pricing.